

CAR BATTERY PREDICTION

Dhiraj · 20 de julio de 2026



Import DEpendenies or library

```
In [120]:
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix, classification_report, accuracy_score
from sklearn.preprocessing import LabelEncoder, MinMaxScaler

import warnings
warnings.filterwarnings('ignore')
```

```
In [121]:
df = pd.read_csv('/content/metadata.csv')
df.head()
```

Out [121]:

	type	start_time	ambient_temperature	battery_id	test_id	uid	filename	Capacity	Re	Rct
0	discharge	[2010. 7. 21. 15. 0. ...	4	B0047	0	1	00001.csv	1.6743047446975208	NaN	NaN
1	impedance	[2010. 7. 21. 16. 53. ...	24	B0047	1	2	00002.csv	NaN	0.05605783343888099	0.20097016584458333
2	charge	[2010. 7. 21. 17. 25. ...	4	B0047	2	3	00003.csv	NaN	NaN	NaN
3	impedance	[2010 7 21 20 31 5]	24	B0047	3	4	00004.csv	NaN	0.05319185850921101	0.16473399914864734
4	discharge	[2.0100e+03 7.0000e+00 2.100...	4	B0047	4	5	00005.csv	1.5243662105099023	NaN	NaN

Data Preprocessing

```
In [122]:
df.shape
```

Out [122]:

(7565, 10)

```
In [123]:
df.info()
```

Out [123]:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7565 entries, 0 to 7564
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   type                   7565 non-null   object
1   start_time             7565 non-null   object
2   ambient_temperature    7565 non-null   int64
3   battery_id             7565 non-null   object
4   test_id                7565 non-null   int64
5   uid                    7565 non-null   int64
6   filename                7565 non-null   object
7   Capacity                2794 non-null   object
8   Re                     1956 non-null   object
9   Rct                     1956 non-null   object
```

```
dtypes: int64(3), object(7)
memory usage: 591.1+ KB
```

```
In [124]:
df.describe()
```

Out [124]:

	ambient_temperature	test_id	uid
count	7565.000000	7565.000000	7565.000000
mean	20.017713	176.012558	3783.000000
std	11.082914	152.174147	2183.971726
min	4.000000	0.000000	1.000000
25%	4.000000	55.000000	1892.000000
50%	24.000000	129.000000	3783.000000
75%	24.000000	255.000000	5674.000000
max	44.000000	615.000000	7565.000000

```
In [125]:
df = df.drop(['start_time', 'battery_id', 'test_id', 'uid', 'filename'], axis = 1)
```

```
In [126]:
df.head()
```

Out [126]:

	type	ambient_temperature	Capacity	Re	Rct
0	discharge	4	1.6743047446975208	NaN	NaN
1	impedance	24	NaN	0.05605783343888099	0.20097016584458333
2	charge	4	NaN	NaN	NaN
3	impedance	24	NaN	0.05319185850921101	0.16473399914864734
4	discharge	4	1.5243662105099023	NaN	NaN

```
In [127]:
df.isnull().sum()
```

Out [127]:

	0
type	0
ambient_temperature	0
Capacity	4771
Re	5609
Rct	5609

```
dtype: int64
```

```
In [128]:
df.info()
```

Out [128]:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7565 entries, 0 to 7564
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   type            7565 non-null   object
1   ambient_temperature  7565 non-null   int64
2   Capacity        2794 non-null   object
3   Re              1956 non-null   object
4   Rct             1956 non-null   object
```

```
dtypes: int64(1), object(4)
memory usage: 295.6+ KB
```

```
In [129]:
df['Capacity'] = pd.to_numeric(df['Capacity'], errors = 'coerce')
```

```
In [130]:
df['Rct'] = pd.to_numeric(df['Rct'], errors = 'coerce')
df['Re'] = pd.to_numeric(df['Re'], errors = 'coerce')
```

```
In [131]:
df.info()
```

```
Out [131]:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7565 entries, 0 to 7564
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   type                  7565 non-null   object
1   ambient_temperature    7565 non-null   int64
2   Capacity              2769 non-null   float64
3   Re                    1947 non-null   float64
4   Rct                   1947 non-null   float64
dtypes: float64(3), int64(1), object(1)
memory usage: 295.6+ KB
```

```
In [132]:
df['Re'] = df['Re'].fillna(df['Re'].mean())
df['Rct'] = df['Rct'].fillna(df['Rct'].mean())
df['Capacity'] = df['Capacity'].fillna(df['Capacity'].mean())
```

```
In [133]:
encoder = LabelEncoder()
```

```
In [134]:
df['type'] = encoder.fit_transform(df['type'])
```

```
In [135]:
df.head()
```

```
Out [135]:
```

	type	ambient_temperature	Capacity	Re	Rct
0	1	4	1.674305	-4.976500e+11	1.055903e+12
1	2	24	1.326543	5.605783e-02	2.009702e-01
2	0	4	1.326543	-4.976500e+11	1.055903e+12
3	2	24	1.326543	5.319186e-02	1.647340e-01
4	1	4	1.524366	-4.976500e+11	1.055903e+12

TRAIN TEST SPLIT

```
In [136]:
X = df.drop('ambient_temperature', axis = 1)
y = df['ambient_temperature']
```

```
In [137]:
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42)
```

```
In [138]:
```

```
scaler = MinMaxScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

MODEL BUILDING

```
In [139]:
model = Sequential()

model.add(Dense(units = 64, activation = 'relu', input_dim = X_train.shape[1]))
model.add(Dropout(0.2))

model.add(Dense(units = 32, activation='relu'))
model.add(Dropout(0.2))

model.add(Dense(units = 32, activation = 'relu'))
model.add(Dropout(0.2))

model.add(Dense(units =1, activation = 'linear'))
```

```
In [140]:
model.compile(optimizer='adam', loss = 'mean_squared_error')
```

```
In [141]:
model.summary()
```

Out [141]:
Model: "sequential_3"

Layer (type)	Output Shape	Param #
dense_9 (Dense)	(None, 64)	320
dropout_6 (Dropout)	(None, 64)	0
dense_10 (Dense)	(None, 32)	2,080
dropout_7 (Dropout)	(None, 32)	0
dense_11 (Dense)	(None, 32)	1,056
dropout_8 (Dropout)	(None, 32)	0
dense_12 (Dense)	(None, 1)	33

Total params: 3,489 (13.63 KB)
Trainable params: 3,489 (13.63 KB)
Non-trainable params: 0 (0.00 B)

```
In [142]:
history = model.fit(X_train , y_train , epochs = 200, batch_size = 32 , validation_data = (X_test , y_test))
```

```
Out [142]:
Epoch 1/200
1/190 [32m] - loss: 383.8862 - val_loss: 109272.8281
Epoch 2/200
1/190 [32m] - loss: 127.9991 - val_loss: 84731.0156
Epoch 3/200
1/190 [32m] - loss: 116.9639 - val_loss: 67778.1406
Epoch 4/200
1/190 [32m] - loss: 116.6087 - val_loss: 54355.0977
Epoch 5/200
1/190 [32m] - loss: 112.7418 - val_loss: 32095.0801
Epoch 6/200
1/190 [32m] - loss: 106.9481 - val_loss: 26564.2188
Epoch 7/200
1/190 [32m] - loss: 109.9540 - val_loss: 19973.5723
```

Epoch 8/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 108.1262 - val_loss: 16165.0420

Epoch 9/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 103.1747 - val_loss: 9148.7773

Epoch 10/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 103.7967 - val_loss: 8238.1504

Epoch 11/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 103.7165 - val_loss: 7463.9888

Epoch 12/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 102.1162 - val_loss: 6014.8633

Epoch 13/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 106.4829 - val_loss: 6954.1851

Epoch 14/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 100.8101 - val_loss: 5917.0781

Epoch 15/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 103.4170 - val_loss: 4342.1470

Epoch 16/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 100.9843 - val_loss: 3043.0525

Epoch 17/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 100.6429 - val_loss: 2722.1650

Epoch 18/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 98.8469 - val_loss: 3455.2278

Epoch 19/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 97.9450 - val_loss: 2894.2329

Epoch 20/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 97.1947 - val_loss: 2730.4119

Epoch 21/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 4ms/step - loss: 95.0307 - val_loss: 1441.1704

Epoch 22/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 93.5501 - val_loss: 2004.4033

Epoch 23/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 93.9089 - val_loss: 1540.1665

Epoch 24/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 95.2353 - val_loss: 1332.1072

Epoch 25/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.4198 - val_loss: 813.8025

Epoch 26/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.5096 - val_loss: 568.6629

Epoch 27/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 96.4194 - val_loss: 356.0292

Epoch 28/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 95.9625 - val_loss: 331.0353

Epoch 29/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.7222 - val_loss: 217.3278

Epoch 30/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 95.4413 - val_loss: 118.4055

Epoch 31/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 97.1943 - val_loss: 105.0114

Epoch 32/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 94.8110 - val_loss: 92.7094

Epoch 33/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.7922 - val_loss: 90.9553

Epoch 34/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.8717 - val_loss: 88.3357

Epoch 35/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.2244 - val_loss: 88.8235

Epoch 36/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 98.4115 - val_loss: 88.5930

Epoch 37/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.3869 - val_loss: 85.3002

Epoch 38/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.7264 - val_loss: 86.1628

Epoch 39/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 94.4408 - val_loss: 86.8562

Epoch 40/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.0988 - val_loss: 85.3269

Epoch 41/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 4ms/step - loss: 95.4045 - val_loss: 85.2202

Epoch 42/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 4ms/step - loss: 97.2133 - val_loss: 90.0253

Epoch 43/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 93.3551 - val_loss: 101.2538

Epoch 44/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.5303 - val_loss: 108.4806

Epoch 45/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.8669 - val_loss: 123.8119

Epoch 46/200

[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.7539 - val_loss: 122.5192

Epoch 47/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 94.2121 - val_loss: 120.1128

Epoch 48/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 94.6976 - val_loss: 112.2375

Epoch 49/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.0430 - val_loss: 131.5302

Epoch 50/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.3718 - val_loss: 114.6952

Epoch 51/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.8186 - val_loss: 104.3098

Epoch 52/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.3010 - val_loss: 108.1553

Epoch 53/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.8009 - val_loss: 90.6649

Epoch 54/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.1037 - val_loss: 86.1713

Epoch 55/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 94.5539 - val_loss: 85.9459

Epoch 56/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.5818 - val_loss: 84.9304

Epoch 57/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.0438 - val_loss: 85.6529

Epoch 58/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.4919 - val_loss: 85.3677

Epoch 59/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 94.2555 - val_loss: 85.8517

Epoch 60/200
[1m190/190][0m [32m-----[0m[37m[0m [1m2s[0m 6ms/step - loss: 92.1168 - val_loss: 86.8281

Epoch 61/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.5792 - val_loss: 89.3645

Epoch 62/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.0795 - val_loss: 99.3272

Epoch 63/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.3253 - val_loss: 92.3837

Epoch 64/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.5907 - val_loss: 93.1669

Epoch 65/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.2644 - val_loss: 87.4385

Epoch 66/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.3677 - val_loss: 91.2504

Epoch 67/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.2269 - val_loss: 86.3750

Epoch 68/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.3304 - val_loss: 85.2115

Epoch 69/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 94.5332 - val_loss: 90.1313

Epoch 70/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.3950 - val_loss: 85.2966

Epoch 71/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.5926 - val_loss: 84.3090

Epoch 72/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.9926 - val_loss: 84.2377

Epoch 73/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.2452 - val_loss: 83.9509

Epoch 74/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 90.5452 - val_loss: 84.6074

Epoch 75/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.6421 - val_loss: 84.2496

Epoch 76/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.2673 - val_loss: 84.7756

Epoch 77/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 4ms/step - loss: 90.7050 - val_loss: 85.1701

Epoch 78/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 92.1139 - val_loss: 84.5396

Epoch 79/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 6ms/step - loss: 93.2477 - val_loss: 83.8815

Epoch 80/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.1670 - val_loss: 84.4404

Epoch 81/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.7087 - val_loss: 84.6049

Epoch 82/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 88.0004 - val_loss: 85.4012

Epoch 83/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 90.1535 - val_loss: 84.5755

Epoch 84/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.2940 - val_loss: 83.8431

Epoch 85/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.1635 - val_loss: 84.3339

Epoch 86/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.3578 - val_loss: 84.5467

Epoch 87/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.1631 - val_loss: 85.2065

Epoch 88/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 92.5570 - val_loss: 84.3801

Epoch 89/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.2495 - val_loss: 84.0494

Epoch 90/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 93.1385 - val_loss: 83.7033

Epoch 91/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.2945 - val_loss: 84.4826

Epoch 92/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 88.8271 - val_loss: 83.9029

Epoch 93/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 84.2024 - val_loss: 84.2907

Epoch 94/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.6880 - val_loss: 84.0507

Epoch 95/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 90.1320 - val_loss: 83.7346

Epoch 96/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 4ms/step - loss: 94.6874 - val_loss: 83.9903

Epoch 97/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 88.9334 - val_loss: 83.5623

Epoch 98/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 89.9807 - val_loss: 86.9603

Epoch 99/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 90.1048 - val_loss: 83.5375

Epoch 100/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.5549 - val_loss: 83.7572

Epoch 101/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 90.2760 - val_loss: 83.7930

Epoch 102/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 88.7449 - val_loss: 84.0333

Epoch 103/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.1518 - val_loss: 83.6807

Epoch 104/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.9723 - val_loss: 84.1466

Epoch 105/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.3342 - val_loss: 83.3253

Epoch 106/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.2453 - val_loss: 83.6926

Epoch 107/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.3832 - val_loss: 83.5858

Epoch 108/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 90.8362 - val_loss: 83.8871

Epoch 109/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.4316 - val_loss: 84.4918

Epoch 110/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 88.7642 - val_loss: 84.1555

Epoch 111/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.6876 - val_loss: 84.3594

Epoch 112/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.2922 - val_loss: 84.0944

Epoch 113/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.6111 - val_loss: 83.9908

Epoch 114/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.3058 - val_loss: 83.4519

Epoch 115/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.4958 - val_loss: 84.4582

Epoch 116/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 90.2425 - val_loss: 83.5879

Epoch 117/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 6ms/step - loss: 86.7147 - val_loss: 83.4666

Epoch 118/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.0082 - val_loss: 83.2603

Epoch 119/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 90.5362 - val_loss: 83.2642

Epoch 120/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 7ms/step - loss: 89.7523 - val_loss: 83.7602

Epoch 121/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 89.3444 - val_loss: 84.7387

Epoch 122/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 85.3866 - val_loss: 83.4594

Epoch 123/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 90.8901 - val_loss: 84.1387

Epoch 124/200
[1m190/190][0m [32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.2344 - val_loss: 83.8217

Epoch 125/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 87.4068 - val_loss: 84.6844

Epoch 126/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 88.0301 - val_loss: 83.7883

Epoch 127/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 85.3790 - val_loss: 83.8099

Epoch 128/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 88.4887 - val_loss: 85.5384

Epoch 129/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 87.3068 - val_loss: 83.6136

Epoch 130/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 84.7699 - val_loss: 83.4055

Epoch 131/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 87.9780 - val_loss: 84.0699

Epoch 132/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 88.4750 - val_loss: 84.0271

Epoch 133/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 5ms/step - loss: 88.4127 - val_loss: 84.2382

Epoch 134/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 5ms/step - loss: 84.5805 - val_loss: 83.8915

Epoch 135/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 5ms/step - loss: 87.7610 - val_loss: 83.8140

Epoch 136/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 85.1284 - val_loss: 83.7031

Epoch 137/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 88.6631 - val_loss: 83.8071

Epoch 138/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 87.3660 - val_loss: 84.4302

Epoch 139/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 88.4608 - val_loss: 83.6368

Epoch 140/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 4ms/step - loss: 88.6025 - val_loss: 83.5654

Epoch 141/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 90.9918 - val_loss: 83.8215

Epoch 142/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 86.4802 - val_loss: 83.7996

Epoch 143/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 86.4620 - val_loss: 83.3642

Epoch 144/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 85.0546 - val_loss: 83.6224

Epoch 145/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 82.5682 - val_loss: 83.2974

Epoch 146/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 84.9589 - val_loss: 84.9459

Epoch 147/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 85.3951 - val_loss: 83.4510

Epoch 148/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 83.3873 - val_loss: 84.3823

Epoch 149/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 85.9254 - val_loss: 83.5619

Epoch 150/200
[1m190/190][0m 32m] [0m 37m][0m 1m2s][0m 5ms/step - loss: 85.9827 - val_loss: 84.5517

Epoch 151/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 5ms/step - loss: 83.5438 - val_loss: 84.9800

Epoch 152/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 90.9592 - val_loss: 85.6616

Epoch 153/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 85.4759 - val_loss: 84.3607

Epoch 154/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 4ms/step - loss: 86.4522 - val_loss: 83.6373

Epoch 155/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 4ms/step - loss: 83.7071 - val_loss: 83.5035

Epoch 156/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 4ms/step - loss: 85.8851 - val_loss: 83.4125

Epoch 157/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 83.9844 - val_loss: 83.6335

Epoch 158/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 85.4996 - val_loss: 84.2293

Epoch 159/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 88.4557 - val_loss: 83.3764

Epoch 160/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 87.0288 - val_loss: 83.3308

Epoch 161/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 89.4437 - val_loss: 84.7099

Epoch 162/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 86.0210 - val_loss: 83.4095

Epoch 163/200
[1m190/190][0m 32m] [0m 37m][0m 1m1s][0m 3ms/step - loss: 86.0487 - val_loss: 83.8177


```

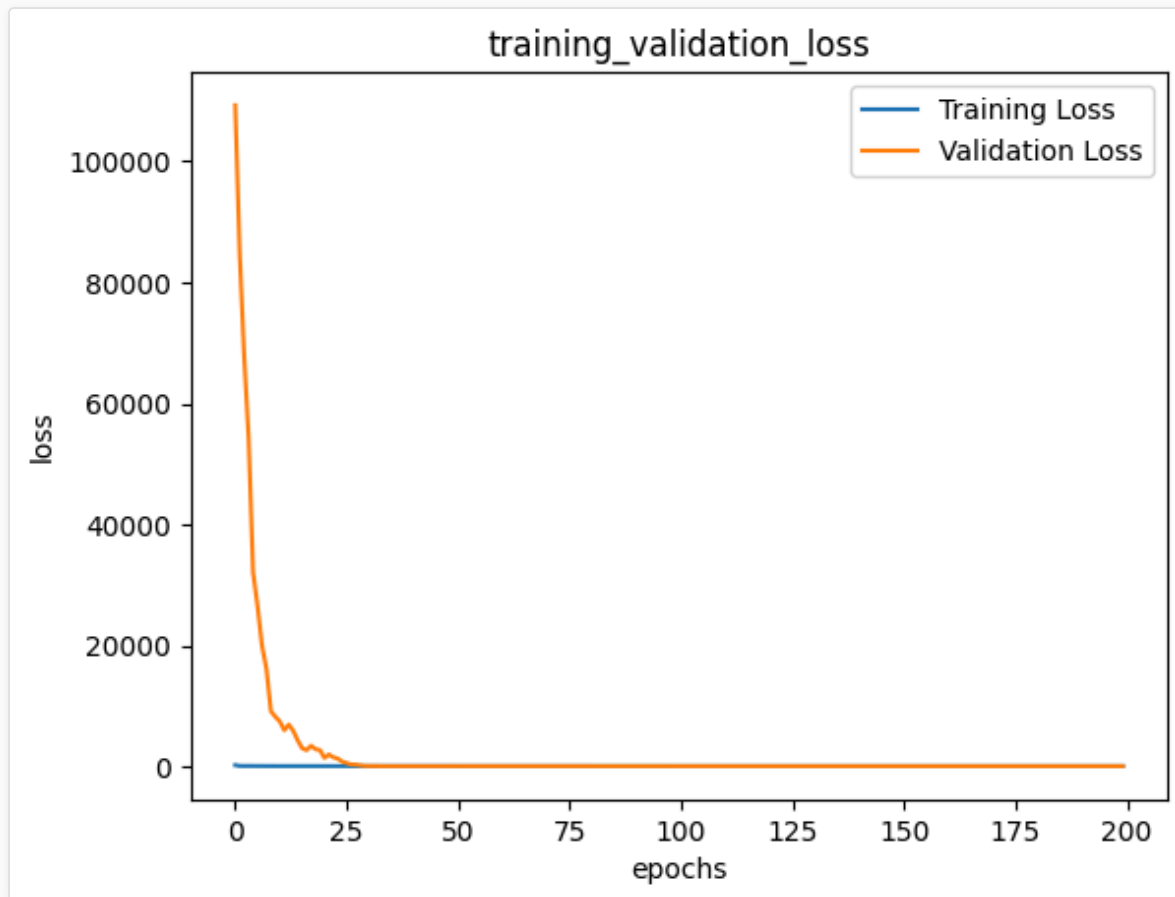
Epoch 164/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 84.3450 - val_loss: 84.2069
Epoch 165/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.1698 - val_loss: 83.5741
Epoch 166/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 81.7230 - val_loss: 83.5556
Epoch 167/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 6ms/step - loss: 87.2744 - val_loss: 83.3243
Epoch 168/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 86.5423 - val_loss: 83.5207
Epoch 169/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 4ms/step - loss: 84.2717 - val_loss: 84.3801
Epoch 170/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.2998 - val_loss: 83.1413
Epoch 171/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 85.3507 - val_loss: 83.7400
Epoch 172/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.0029 - val_loss: 83.4911
Epoch 173/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 83.2934 - val_loss: 83.4395
Epoch 174/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 84.4798 - val_loss: 83.3825
Epoch 175/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 88.5863 - val_loss: 83.7554
Epoch 176/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 85.5875 - val_loss: 83.9115
Epoch 177/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.7402 - val_loss: 85.0384
Epoch 178/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.4875 - val_loss: 83.3187
Epoch 179/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 85.1169 - val_loss: 83.4655
Epoch 180/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.8870 - val_loss: 83.5905
Epoch 181/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 87.2270 - val_loss: 85.6903
Epoch 182/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 88.0341 - val_loss: 84.0777
Epoch 183/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 84.8798 - val_loss: 83.6372
Epoch 184/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 88.6018 - val_loss: 84.8956
Epoch 185/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.0072 - val_loss: 84.9762
Epoch 186/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 4ms/step - loss: 88.9217 - val_loss: 83.9155
Epoch 187/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 5ms/step - loss: 85.7765 - val_loss: 85.1152
Epoch 188/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 4ms/step - loss: 84.5476 - val_loss: 83.6231
Epoch 189/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.7029 - val_loss: 83.5686
Epoch 190/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 85.8183 - val_loss: 83.3665
Epoch 191/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.9459 - val_loss: 83.5614
Epoch 192/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.7000 - val_loss: 83.6134
Epoch 193/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 85.8526 - val_loss: 83.4347
Epoch 194/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 85.5970 - val_loss: 84.2022
Epoch 195/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.1916 - val_loss: 85.0392
Epoch 196/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 91.4064 - val_loss: 83.5327
Epoch 197/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 86.7448 - val_loss: 83.4434
Epoch 198/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 83.6176 - val_loss: 85.1675
Epoch 199/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 83.5466 - val_loss: 84.1477
Epoch 200/200
[1m190/190][0m 32m-----[0m[37m[0m [1m1s[0m 3ms/step - loss: 83.9599 - val_loss: 83.5425

```

Model Evolution

```
In [145]:
plt.plot(history.history['loss'], label = 'Training Loss')
plt.plot(history.history['val_loss'], label = 'Validation Loss')
plt.legend()
plt.xlabel('epochs')
plt.ylabel('loss')
plt.title('training_validation_loss')
plt.show()
```

Out [145]:



```
In [146]:
model.evaluate(X_test, y_test)
```

```
Out [146]:
1/48 [1m48s/48m 0m 32m 0m 0m 37m 0m 1m0s 0m 2ms/step - loss: 87.8329
83.54247283935547
```

SAVE MODEL

```
In [147]:
import joblib

# Save model
model.save('battery_temp_model.h5')

# Save scaler and encoder
joblib.dump(scaler, 'scaler.pkl')
joblib.dump(encoder, 'encoder.pkl')

print("All saved successfully!")
```

```
Out [147]:
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is
considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my_model.keras')` or
```

```
`keras.saving.save_model(model, 'my_model.keras')`.  
All saved successfully!
```